



Based in Grenoble, France
US Citizen, VLS-TS Resident Visa

Katelyn M. Schoedl

Electrical Engineer and Research Coordinator

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Education

Bachelor of Science in Electrical Engineering

University of Illinois at Urbana-Champaign

August 2015 – May 2019 (Bac+4)

Illinois, USA

- Coursework: Computer Programming, Electromagnetics, Semiconductor Physics, Analog and Digital Circuits
- Internships: GE Global Research (Summer 2016 & 2017), Amazon (Summer 2018)
- Exchange Semester: Technical University of Denmark (Autumn 2017)

Professional Experience

Research Coordinator (Photonics)

University of Washington, College of the Environment, Dept. of Earth & Space Sciences

May 2024 – May 2025

Seattle, WA

- Operated and deployed fiber optic interrogator systems (Sintella Onyx, ASN OptoDAS, FEBUS G1-C) for DAS/DTS/DSS sensing modes across seismic, ocean acoustic, and glacier/ice-ocean dynamics applications.
- Conducted field deployments across six sites including offshore fiber installations in Homer AK, San Juan Islands WA, and Morro Bay CA, and seismic monitoring deployments at Mt. Rainier and Alpentel Ski Resort, WA; handled all pre-deployment testing, independent equipment transport, live-buffer drive swaps, flight, ferry and vehicle logistics.
- Maintained daily operational oversight of concurrent DAS deployments across domestic and international field sites, managing data acquisition systems, remote monitoring infrastructure, and storage logistics across petabyte-scale datasets.
- Configured remote network access for deployed interrogator systems, coordinated RAID storage architecture decisions with university IT, redesigned web presence using WordPress and JavaScript, and ensured data handling compliance including federal data scrubbing protocols.
- Coordinated international equipment shipments and customs carnets for Antarctic and Greenland deployments; managed procurement and lab inventory; facilitated dataset publication through DOI creation with UW Libraries; collaborated closely with USGS, EarthScope, and PNSN; contributed to publications and credited in Shi et al. 2025, Seismological Research Letters.
- Actively participated in research group colloquia and engaged in future planning discussions; contributed to signal analysis including data cross-correlation and live characterization datasets.
- BOAT Ocean Acoustics Workshop for applied signal processing concepts, Seattle, Washington (2025).

Hardware Engineer II (Signal Integrity)

Microsoft, Cloud AI Hardware and Advanced Signal Engineering

September 2021 – September 2023

Seattle, WA

- Performed passive and active signal integrity characterization of PCIe and DDR5 traces across FPGA die, custom silicon packaging, connectors, and board-level interconnects using differential probing, VNA S-parameter analysis, and time-domain eye-diagram methods.
- Identified component-level signal integrity defects through comparative validation of boards across manufacturers and production origins, enabling root cause isolation and cost savings in hardware qualification.
- Supported stackup design decisions alongside motherboard EEs, contributing to material selection and layer thickness tradeoffs that determined high-speed signal propagation behavior.
- Built end-to-end simulation models of signal paths from silicon to GPU and PCIe endpoints using ANSYS HFSS and Keysight ADS, analyzing microband and geometry effects on transmission performance.
- DesignCon communications design conference attendee, Santa Clara, California (2022).

Hardware Design Engineer (Satellite Bus Avionics)

Amazon, Project Kuiper

April 2020 – September 2021

Seattle, WA

- Owned end-to-end hardware development for magnetometer and sun sensor subsystems during early-phase pre-PDR program development (prior to launch provider selection and during initial lab standup) including electrical, mechanical, and functional requirements definition, FOV analysis, and ConOps documentation.
- Designed and populated custom test boards for heavy-ion single-event effects (SEE) characterization at the Lawrence Berkeley National Laboratory 88-Inch Cyclotron; prepared delidded components, coordinated beamline exposure parameters, and performed pre- and post-irradiation functional testing of semiconductor devices; contributed to Kuiper's internal COTS component radiation qualification library.
- Led component selection, schematic design, and PCB layout for avionics sensor prototypes; coordinated with contract manufacturers on assembly requirements and parts library verification.
- Performed sensor bring-up, functional validation, and photodiode characterization using test tools such as sun emulator, mechanical test bed, and Helmholtz coil; conducted hardware prototype field validation.
- Supported PDR and CDR design reviews and contributed to cross-team integration reviews.
- Maintained comprehensive engineering documentation across requirements, design decisions, test results, and manufacturing handoff throughout the full hardware development lifecycle.

Skills

Language

English (native)
French, Spanish (A2)

Instrumentation

Photonics: DAS/DTS/DSS, Optical Connectors & Fiber Components, Fusion Splicing, Interrogators (Sintella Onyx, ASN OptoDAS, FEBUS G1-C)

Electronic Test & Measurement: VNA, S-parameters, Eye-diagram Analysis, Oscilloscope, Passive & Active Probing, Photodetector Characterization, EMI/EMC Validation, Thermal/Mechanical Environmental Qualification

Simulation and Design

Hardware Design: Schematic, PCB Layout, Board Bring-up, Stackup Design, PCB Fabrication & Assembly Processes

Simulation & Analysis: ANSYS HFSS, Keysight ADS, Signal Integrity Analysis, Channel Simulation, Noise Characterization

Programming and Development

Languages & Tools: Python, MATLAB, C/C++, JavaScript, LaTeX, Git, Linux/Bash, Docker, Jupyter

Infrastructure: RAID Storage Architecture, Data Acquisition Pipelines, Remote System Monitoring

Logistics and Operations

Logistics: International Shipping, Customs & Carnets, Lab Budget & Procurement, Inventory Management, Project Tracking & Milestone Management, Facilities Booking

Research Operations: Federal Data Compliance, Petabyte-Scale Data Management, Grant Management & Documentation, Technical Reporting, Multi-institution & International Research Collaboration

Field Activities

Training: AIARE Avalanche and Rescue, Wilderness First Responder (WFR) with AED and CPR Certification, OSHA 10 Safety Training

Affiliations: AMGA Professional Member, American Alpine Club, Club Alpin Français, SnowGoat Skimo Volunteer, WA State Rare Plant Monitor